

Can a small robot AI beat the big ones?

A 67-idea search, and what survived.

We ran an automated research lab for a day: dozens of AI agents mined ideas from 14 scientific fields, checked each against published work, and attacked the survivors the way harsh journal reviewers would. The honest result: **no idea survived unwounded** — and that finding, plus one repair-minded winner with a free first experiment, is the deliverable.

67

candidate mechanisms
generated & screened

4

passed all six review gates

0

survived adversarial debate
intact

R4

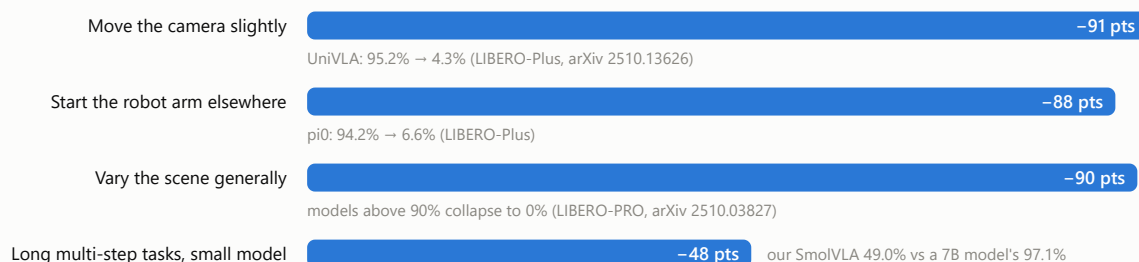
committee's winner —
“check & retry”
(confidence 0.55)

1 · The goal, in plain terms

A **VLA** (vision-language-action model) is an AI that looks at camera images, reads an instruction like “*put the bowl in the drawer*”, and outputs robot arm movements. Big VLAs (7 billion parameters) do this well. Our lab's model, **SmolVLA**, is 15× smaller (450 million) — cheap and fast, but it completes only **49%** of long multi-step tasks in the standard simulated benchmark (LIBERO-Long), where the big models reach ~97%. The mission: find a **genuinely new** add-on mechanism — never tried in robotics AI — that closes part of that gap without retraining the big “brain” (the backbone stays frozen), publishable in a top journal.

2 · What is actually broken in today's robot AIs

Documented failures from 2025–26 benchmark studies. The bars show how many percentage points of success rate vanish when one thing changes. These are the targets any new idea must hit.



Success-rate drop, percentage points. Every figure traces to a fetched source or our own logged run — nothing estimated.

The pattern that matters: today's models don't fail because they “think” wrongly — they fail because **small changes to what the camera sees scramble their perception**, and because on long tasks **one silent slip early on ruins everything after it**.

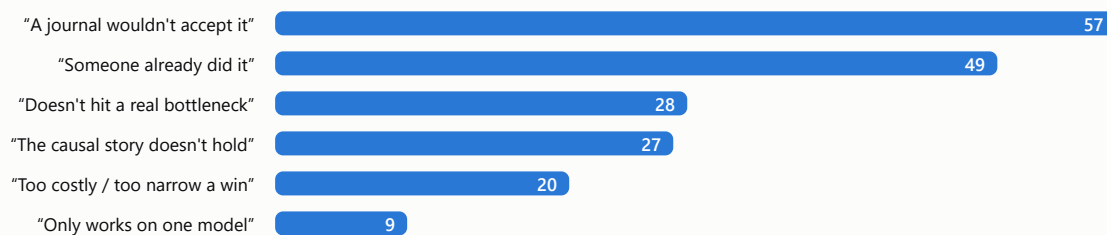
3 · How the search worked

Fourteen scanner agents mined mechanisms from fields that have never touched robot AI — control theory, neuroscience, information theory, computer architecture, databases, immunology, networking. Every surviving idea then had to pass **six gates** (Is it new? Does it hit a real problem? Is the reasoning sound? Does it generalize? Is it efficient? Would a journal accept it?) and finally **three rounds of adversarial debate** where reviewer-agents tried to kill it. When everything died, two independent “surgeon” agents engineered 8 revised ideas specifically to answer the recorded attacks — and those went through the same gauntlet.



The filter was deliberately as harsh as a top journal's review process. 4 ideas that were audit-killed with strong evidence never reached the gates.

4 · Where ideas died — and the lessons that generalize



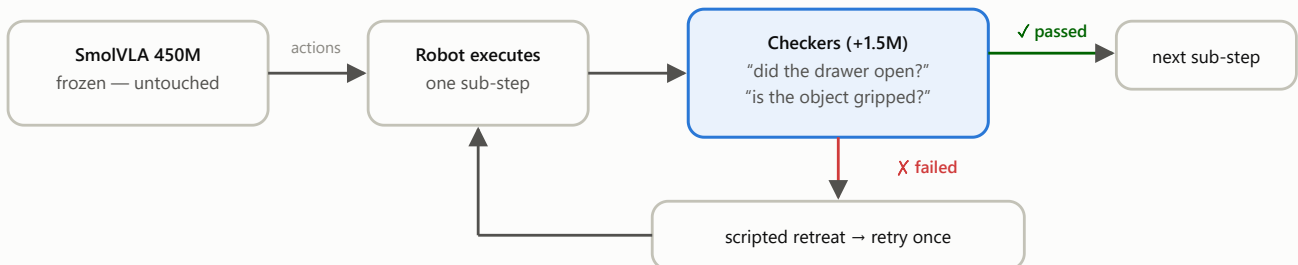
Number of candidates failing each gate (one idea can fail several), out of 63 gated.

Five lessons the graveyard teaches (each verified across multiple kills)

1. **You can't fix broken perception after the fact.** Every monitor, safety gate or corrector that sits *behind* the frozen vision system died the same death: the shifted camera corrupts its inputs too.
2. **Fixes that need “example perturbations” collapse into data augmentation.** If your fix trains on varied lighting/views, then simply training the policy on that same data is a fair baseline — and it usually captures the gain.
3. **One benchmark win counted three ways is still one win.** Ideas claiming success + robustness + generalization from the same measurement failed the multi-benefit requirement.
4. **Borrowed formalisms must not degenerate.** Fancy machinery (evidence fusion, coding theory, database transactions) often reduces to a trivial gate or heuristic once instantiated on a robot policy.
5. **The obvious idea-space is genuinely mined out (2025–26).** Test-time search, action tokenizers, replan schedulers, memory modules, adapters — all have named recent papers within one thin twist.

5 · The winner: “Check & Retry” (R4 — Declarative Postcondition Retry)

The idea in one sentence: after each sub-step, tiny add-on “checkers” (1.5M parameters — 0.3% extra) verify from the robot's own features that the sub-goal *physically happened* — did the drawer actually open? is the object actually in the gripper? — and if not, the robot **backs up and tries that step again** instead of blindly continuing.



Why it survived where 66 died: the retry changes the WORLD (the robot physically resets the situation) — it never has to trust corrupted perception twice in the same way.

Why the committee picked it (3 independent judges + chair, confidence 0.55): long tasks fail multiplicatively — ten 93%-reliable steps compound to ~49%. One retry lifts each step's odds from p to $p(2-p)$ (e.g. $0.93 \rightarrow 0.995$), which is exactly the structure of our measured gap. Expected gain if the premise holds: **+5 to +15 points on LIBERO-Long**. All three scoring lenses gave it the top expected-improvement score.

Its honest weakness: “verify then retry” is classical robotics — the novelty gate failed it. So it publishes as a rigorous *measurement + mechanism* paper (a quantified failure taxonomy of a small VLA, plus the fix), targeted at **RA-L or TMLR** rather than a “new mechanism” headline at CoRL.

The first experiment is free: re-label our existing 100 failed runs. If fewer than 25% failed because of a silent unnoticed sub-step failure, the idea dies for \$0 — and the failure taxonomy still publishes.

6 · The final top 10

■ Engineered this run (revision round) ■ Mined from the 14-field scan



Committee average across 7 scoring axes (novelty, expected gains, difficulty inverted, publishability, impact). † = chair demoted below arithmetic order for a load-bearing flaw. Full per-judge scores in the artifacts folder.

Runner-up · R1

The **novelty lottery ticket**: re-calibrate perception once per episode using the first glance. Highest novelty score (5/5/5) but its core premise is likelier false than true — a ~1-day experiment decides. If it survives, it becomes the paper's headline and R4 the second contribution.

Runner-up · S3

Learn a new task from 10 demos **without changing any weights** (kernel lookup + knowing-when-to-abstain). Well-posed and cheap to falsify; ceiling: its parts are known techniques, so it supports rather than leads a paper.

7 · The plan: kill cheaply, build only what survives

Every step has a pre-registered “kill criterion” decided before it runs — the project's standing rule. All compute on the lab's A100 GPU (verified free).

Step	What happens	GPU-hours	Dies immediately if...
0a	R4 free test — re-label the 100 existing failure logs	0	<25% of failures are silent sub-step slips
0b	R3 side-experiment — decompose baseline randomness (valuable either way)	2	—
1	Train R4's checker probes on cached features	2	pose error > 1cm, or >3% false alarms
2	R4 pilot — retreat-and-retry on 3 long tasks	12	a retry isn't better than a fresh attempt
3	R1 premise test — is the perception shift repairable at all?	18–24	shift lies outside the learned repair space (~60% likely)
4	S3 premise test — offline, on cached features	1–2	lookup reconstruction is 1.5× worse than fine-tuning
5	Full build of survivors — 500 runs × 3 seeds, honest matched-budget scoring	40–80	standard statistics decide

Likely total: ~20–40 GPU-hours if only R4 survives (≈2–3 days); ~80–120 if all three do (≈1–2 weeks on the shared machine).

8 · Honesty box: risks & confidence

Risk	Likelihood	What happens then
R4's free test kills the premise (failures are mostly early misperception, not silent slips)	~50%	The failure taxonomy still publishes as an honest measurement paper; the run's five lessons (§4) stand.
Retries eat the time budget and reviewers object	medium	Pre-registered matched-budget comparison already in the plan.
Someone publishes VLA verify-and-retry first	medium	Move fast; our unique asset is the reproduced-baseline failure taxonomy.

Committee confidence in the winner: 0.55. Confidence this run produced a *publishable research program* (winner + fallback + verified negative findings): **medium-high**. Confidence any of this beats 7-billion-parameter models as a headline: **low** — that race is already lost to 2025's literature; the winnable race is reliability per parameter, honestly measured.

Process: 7 orchestrated multi-agent workflows + 2 independent surgeon agents (~60 sub-agents; Sonnet workers, Fable-5 judges, Opus fallback) · every kill/keep/rank decision made by ≥3 independent agents · every number traced to a run log or fetched source · full machine-readable record in `reports/mission-artifacts-2026-07-10/` · companion technical report: `reports/vla-novelty-mission-2026-07-10.md`